Final Project

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1. Project Background

1.1. Introduction

Rice (Oryza sativa L.) is a major staple food crop in Nepal and plays a key role in food security and rural livelihoods. Improving agronomic practices is essential to increase productivity and economic returns.

One important management practice is seedling density per hill, which can influence crop growth, yield, and profitability.

1.2. Objective

The objective of this study is to evaluate the effect of different seedling densities (1–5 seedlings per hill) on growth, grain yield, and economic performance of spring rice.

1.3. Research Question

How does seedling density affect growth, grain yield, and economic return of spring rice?

1.4. Experimental Design

The experiment was conducted using a Randomized Complete Block Design (RCBD) with: - 5 treatments (seedling densities: 1, 2, 3, 4, 5 seedlings per hill) - 4 replications Each treatment was randomly assigned within each replication.

1.5. Dataset Description

The dataset contains 20 observations collected from a field experiment. The variables include:

- rep: replication/block
- trt: treatment (seedling density)
- ph: plant height (cm)
- nt: number of tillers per square meter

- et: effective tillers per square meter
- pl: panicle length (cm)
- fg: filled grains per panicle
- tgw: thousand grain weight
- sp: sterility percentage
- gy: grain yield (tons/ha)
- sy: straw yield (tons/ha)
- hi: harvest index
- gr: gross return (Nepali Rupees per ha)
- nr: net return (Nepali Rupees per ha)
- bc: benefit-cost ratio

1.6. Data Import and Preparation

```r
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.5.2
```

```r
library(readr)
```

```
## Warning: package 'readr' was built under R version 4.5.2
```

```r
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.5.2

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.5.2

## Warning: package 'tibble' was built under R version 4.5.2

## Warning: package 'tidyr' was built under R version 4.5.2

## Warning: package 'purrr' was built under R version 4.5.2

## Warning: package 'stringr' was built under R version 4.5.2

## Warning: package 'forcats' was built under R version 4.5.2

## Warning: package 'lubridate' was built under R version 4.5.2

## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v forcats   1.0.1     v stringr   1.6.0
## v lubridate 1.9.4     v tibble    3.3.1
## v purrr     1.2.1     v tidyr     1.3.2

## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
data <- read.csv("ProjectData.csv")

data <- data %>%
  mutate(rep = as.factor(rep),trt = as.factor(trt))

str(data)
```

```
## 'data.frame':    20 obs. of  15 variables:
##  $ rep: Factor w/ 4 levels "1","2","3","4": 1 1 1 1 1 2 2 2 2 2 ...
##  $ trt: Factor w/ 5 levels "1","2","3","4",..: 1 2 3 4 5 1 2 3 4 5 ...
##  $ ph : num  95.6 95 90.7 90.2 88.8 ...
##  $ nt : int  290 380 260 285 335 250 325 335 265 290 ...
##  $ et : num  222 318 278 238 228 ...
##  $ pl : num  24.6 23.7 23.6 23 23.9 ...
##  $ fg : num  106 122 119 100 108 ...
##  $ tgw: num  21.9 17.1 21.1 24 22.3 ...
##  $ sp : num  23 25 29 29 28.5 ...
##  $ gy : num  5.36 6.87 6.6 5.55 5.19 6.27 6.27 5.89 5.08 4.9 ...
##  $ sy : num  9.1 10.2 8.7 8.44 8.9 7.01 10.8 9.6 8.26 8.8 ...
##  $ hi : num  0.371 0.402 0.431 0.397 0.368 ...
##  $ gr : int  144720 185490 178200 149850 140130 169290 169290 159030 137160 132300 ...
##  $ nr : int  73882 113772 105602 76372 65772 98452 97572 86432 63682 57942 ...
##  $ bc : num  2.04 2.59 2.45 2.04 1.88 ...
```

```r
summary(data)
```

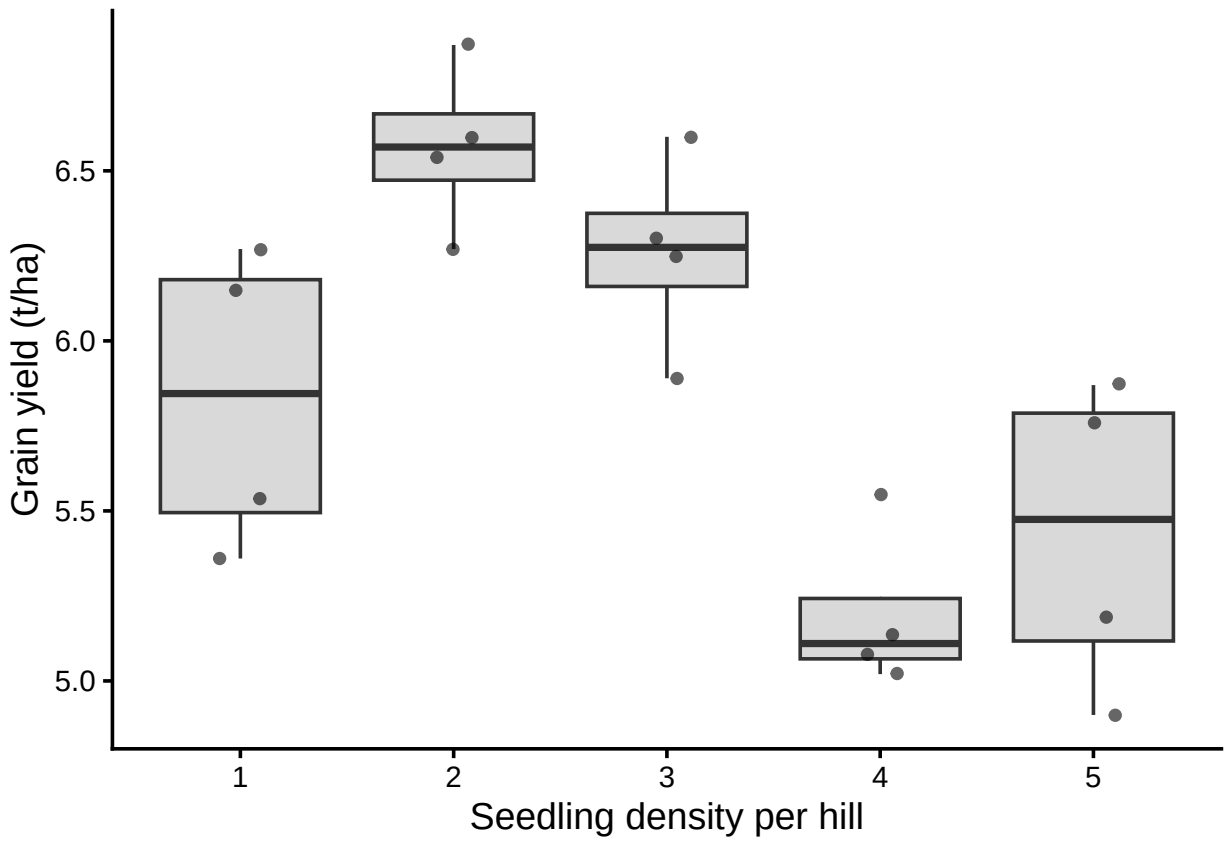
```
##  rep   trt         ph              nt              et              pl
##  1:5   1:4   Min.   :87.10   Min.   :245.0   Min.   :215.0   Min.   :22.30
##  2:5   2:4   1st Qu.:88.62   1st Qu.:282.5   1st Qu.:226.2   1st Qu.:23.02
##  3:5   3:4   Median :90.90   Median :305.0   Median :253.8   Median :23.71
##  4:5   4:4   Mean   :91.14   Mean   :305.5   Mean   :253.9   Mean   :23.59
##        5:4   3rd Qu.:93.25   3rd Qu.:327.5   3rd Qu.:268.1   3rd Qu.:24.06
##              Max.   :95.94   Max.   :380.0   Max.   :317.5   Max.   :24.75
##        fg             tgw              sp              gy
##  Min.   :100.4   Min.   :17.07   Min.   :22.43   Min.   :4.900
##  1st Qu.:108.8   1st Qu.:19.87   1st Qu.:24.41   1st Qu.:5.317
##  Median :116.5   Median :21.84   Median :26.59   Median :5.880
##  Mean   :114.0   Mean   :21.40   Mean   :26.36   Mean   :5.857
##  3rd Qu.:119.0   3rd Qu.:22.43   3rd Qu.:28.62   3rd Qu.:6.277
##  Max.   :122.4   Max.   :24.62   Max.   :29.98   Max.   :6.870
##        sy               hi               gr               nr
##  Min.   : 7.010   Min.   :0.3577   Min.   :132300   Min.   : 57942
##  1st Qu.: 8.395   1st Qu.:0.3726   1st Qu.:143573   1st Qu.: 71855
##  Median : 8.850   Median :0.3842   Median :158760   Median : 85282
##  Mean   : 9.008   Mean   :0.3944   Mean   :158153   Mean   : 85555
##  3rd Qu.: 9.637   3rd Qu.:0.4101   3rd Qu.:169493   3rd Qu.: 97792
##  Max.   :10.800   Max.   :0.4721   Max.   :185490   Max.   :113772
##        bc
##  Min.   :1.779
##  1st Qu.:2.002
##  Median :2.161
##  Mean   :2.181
##  3rd Qu.:2.368
##  Max.   :2.586
```

2. Data Visualization

Data visualization was carried out using ggplot2 to explore treatment effects and relationships among agronomic and economic variables. Figures were prepared with consistent themes, clear labeling, and minimal visual clutter.

2.1. Effect of Seedling Density on Grain Yield

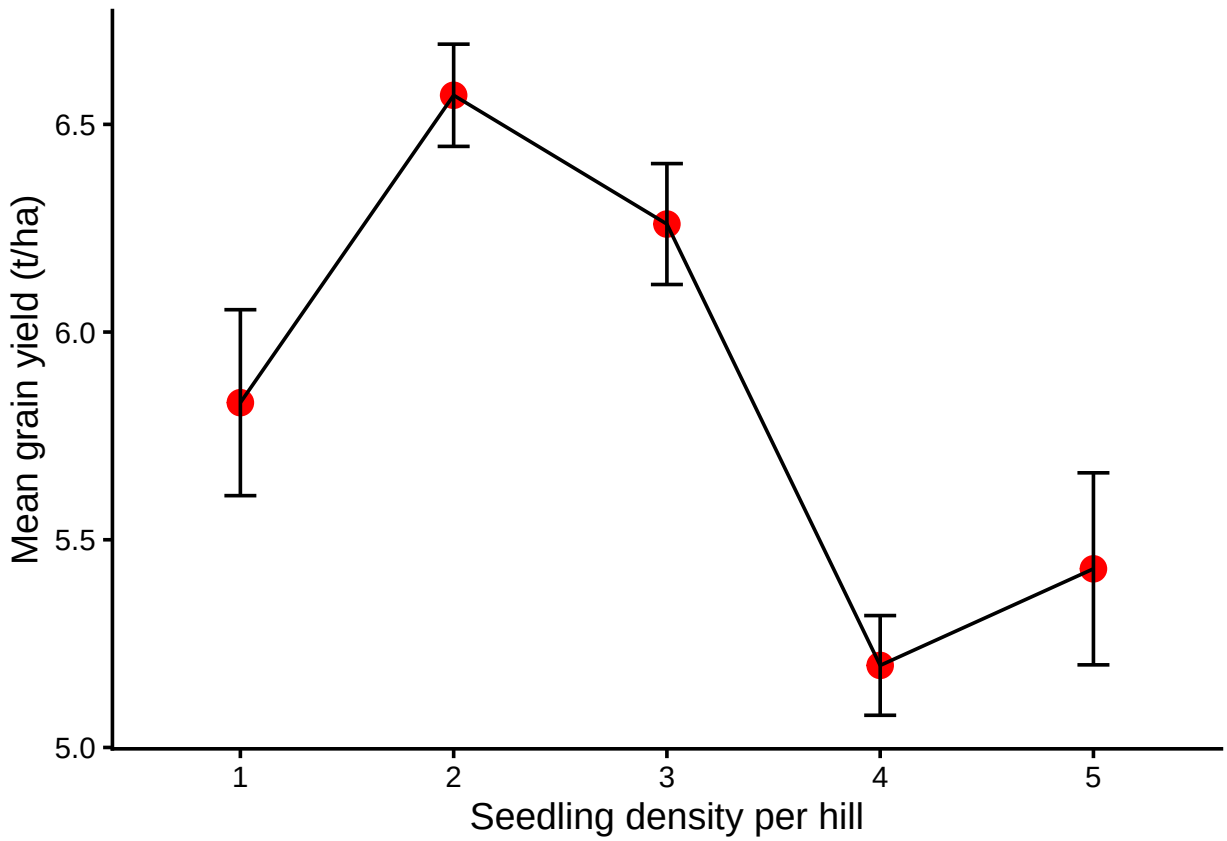
```r
ggplot(data, aes(x = trt, y = gy)) +
  geom_boxplot(fill = "grey85", outlier.shape = NA) +
  geom_jitter(width = 0.15, alpha = 0.6, size = 1.5) +
  labs(x = "Seedling density per hill",
       y = "Grain yield (t/ha)") +
  theme_classic(base_size = 14)
```



This figure shows the distribution of grain yield across seedling density treatments. It highlights both variability and central tendency within each treatment.

2.2. Treatment Mean Response (Yield Trend)

```
ggplot(data, aes(x = trt, y = gy)) +
  stat_summary(fun = mean, geom = "point", size = 4, color = "red") +
  stat_summary(fun = mean, geom = "line", aes(group = 1)) +
  stat_summary(fun.data = mean_se, geom = "errorbar", width = 0.15) +
  labs(x = "Seedling density per hill",
       y = "Mean grain yield (t/ha)") +
  theme_classic(base_size = 14)
```

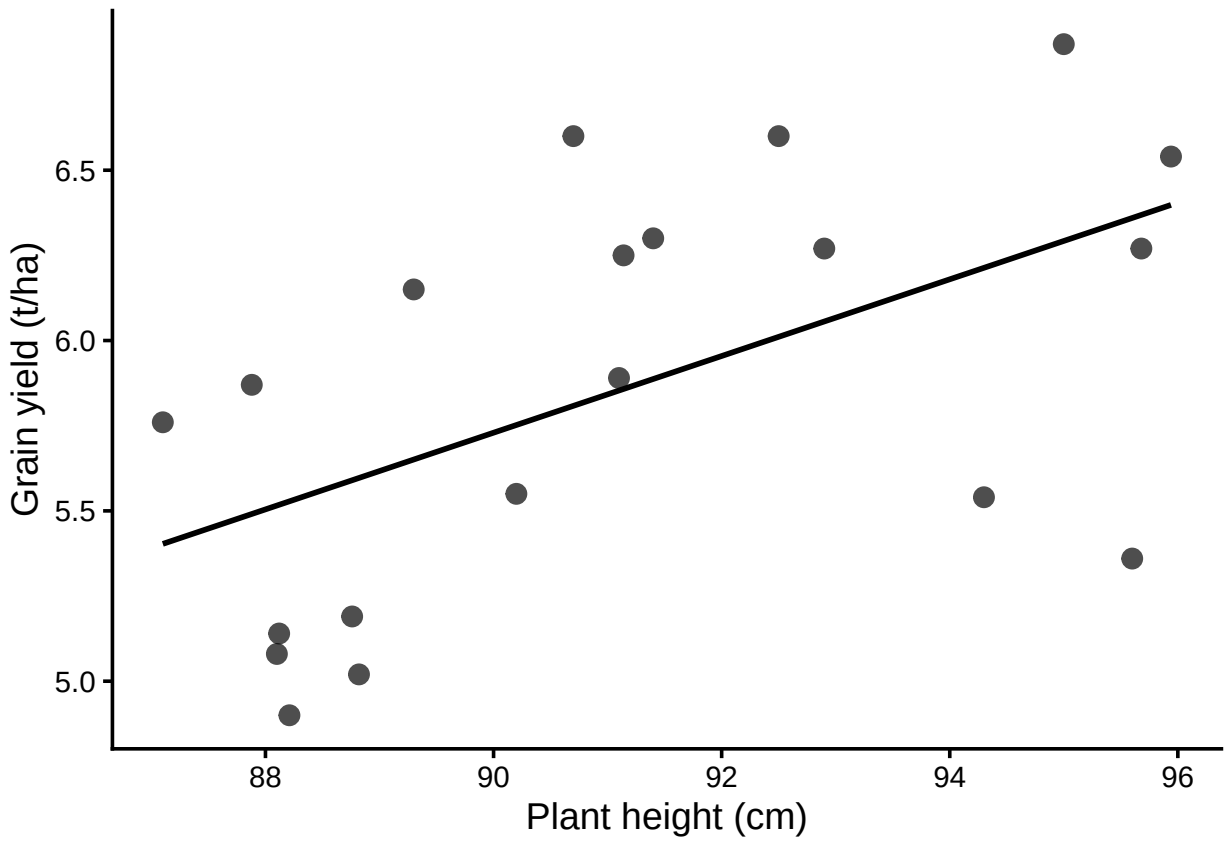


This figure presents mean grain yield across treatments with standard error bars, allowing comparison of productivity trends across seedling densities.

2.3. Plant Height and Grain Yield Relationship

```
ggplot(data, aes(x = ph, y = gy)) +
  geom_point(size = 3, alpha = 0.7) +
  geom_smooth(method = "lm", se = FALSE, color = "black", linewidth = 1) +
  labs(x = "Plant height (cm)",
       y = "Grain yield (t/ha)") +
  theme_classic(base_size = 14)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

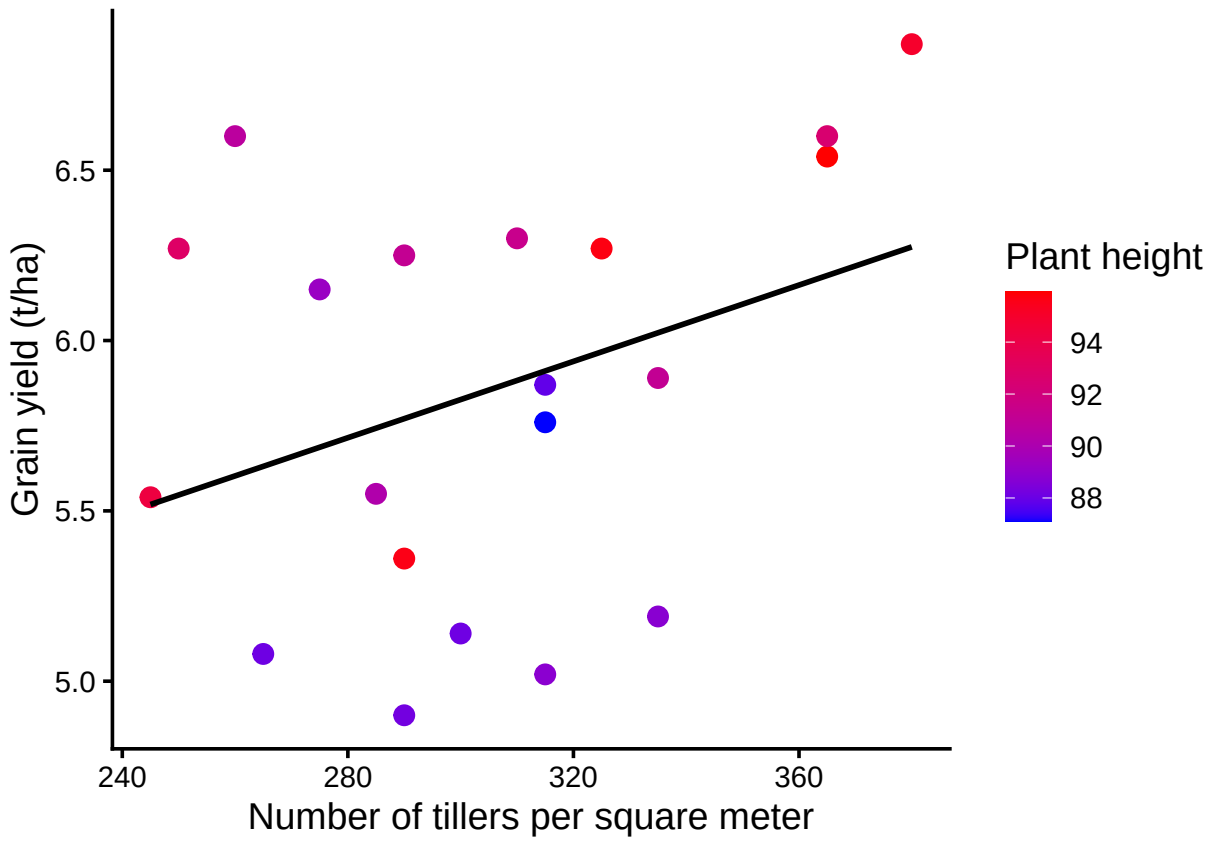


This figure shows the relationship between plant height and grain yield, indicating how vegetative growth contributes to yield formation.

2.4. Tillering Effect on Grain Yield (Multi-variable plot)

```
ggplot(data, aes(x = nt, y = gy, color = ph)) +
  geom_point(size = 3) +
  geom_smooth(method = "lm", se = FALSE, color = "black", linewidth = 1) +
  scale_color_gradient(low = "blue", high = "red") +
  labs(x = "Number of tillers per square meter",
       y = "Grain yield (t/ha)",
       color = "Plant height") +
  theme_classic(base_size = 14)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

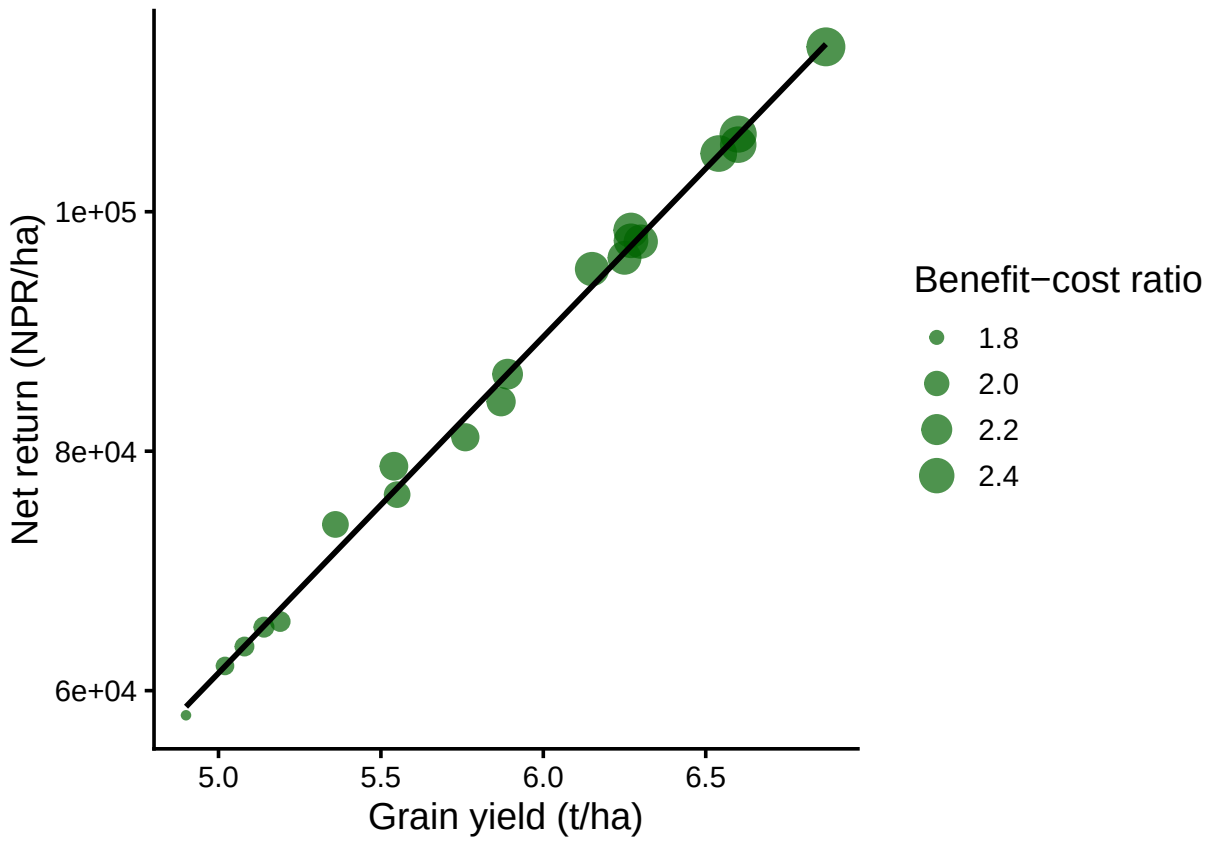


This figure illustrates how tiller density and plant height jointly influence grain yield, highlighting multiple yield-contributing traits.

2.5. Yield and Economic Return Relationship

```
ggplot(data, aes(x = gy, y = nr)) +
  geom_point(aes(size = bc), alpha = 0.7, color = "darkgreen") +
  geom_smooth(method = "lm", se = FALSE, color = "black", linewidth = 1) +
  labs(x = "Grain yield (t/ha)",
       y = "Net return (NPR/ha)",
       size = "Benefit-cost ratio") +
  theme_classic(base_size = 14)
```

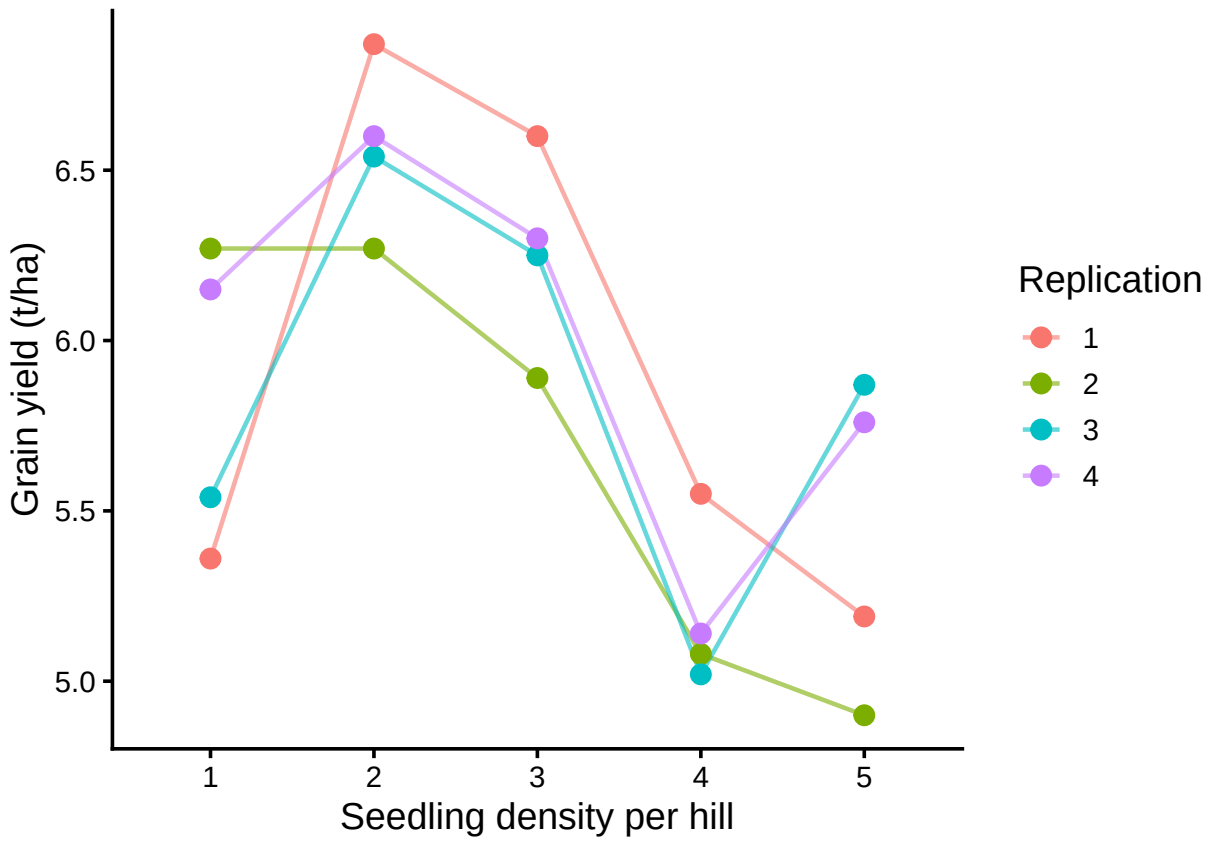
```
## `geom_smooth()` using formula = 'y ~ x'
```



This relationship shows that higher grain yield is associated with increased net returns, indicating improved economic efficiency.

2.6. Yield Stability Across Replications (RCBD Structure)

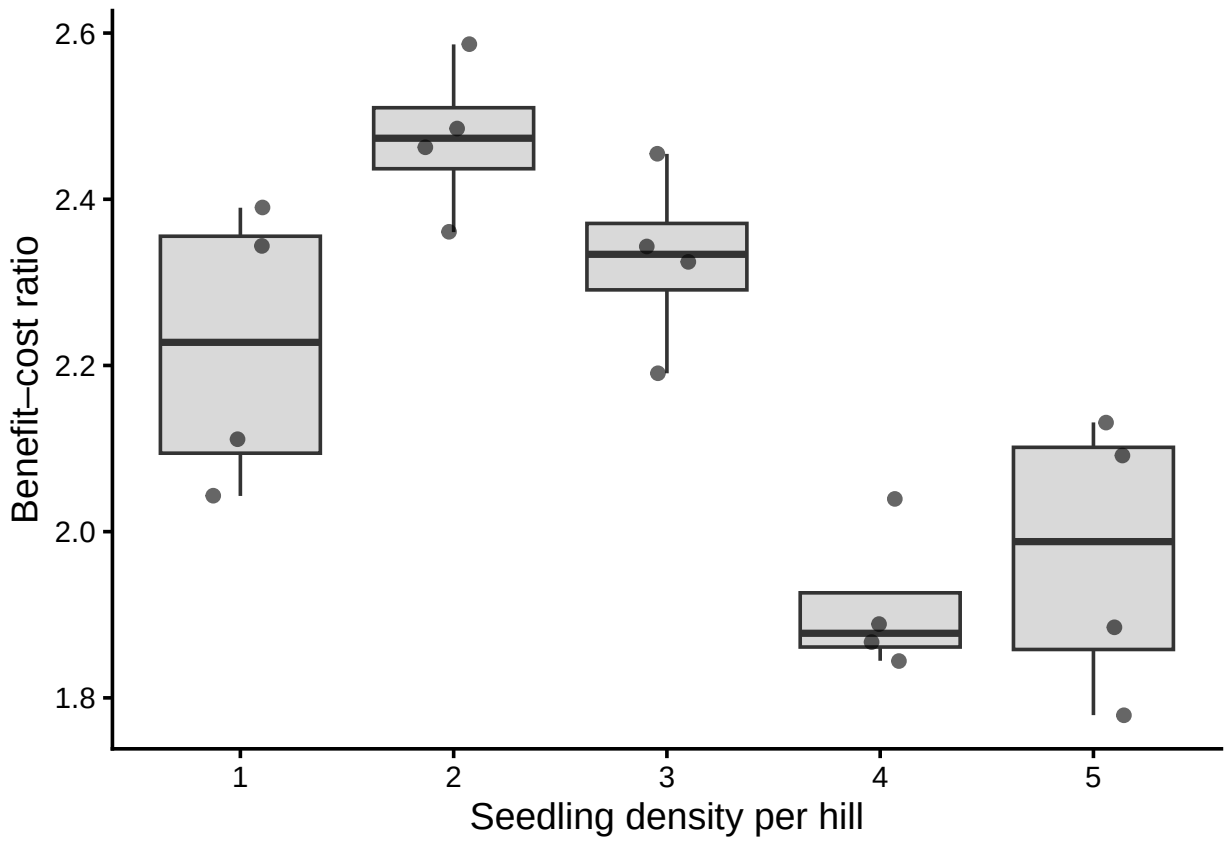
```
ggplot(data, aes(x = trt, y = gy, group = rep, color = rep)) +
  geom_line(linewidth = 0.8, alpha = 0.6) +
  geom_point(size = 3) +
  labs(x = "Seedling density per hill",
       y = "Grain yield (t/ha)",
       color = "Replication") +
  theme_classic(base_size = 14)
```



This figure illustrates variation in treatment response across replications, reflecting the structure of the randomized complete block design.

2.7. Economic Efficiency (Benefit–Cost Ratio)

```
ggplot(data, aes(x = trt, y = bc)) +
  geom_boxplot(fill = "grey85", outlier.shape = NA) +
  geom_jitter(width = 0.15, alpha = 0.6) +
  labs(x = "Seedling density per hill",
       y = "Benefit-cost ratio") +
  theme_classic(base_size = 14)
```



This figure compares economic efficiency across seedling density treatments, showing differences in profitability among management practices.

3. Statistical Analysis

3.1. Effect of Seedling Density on Grain Yield

```
model_gy <- lm(gy ~ trt, data = data)
summary(model_gy)
```

```
##
## Call:
## lm(formula = gy ~ trt, data = data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -0.5300 -0.2525 -0.0200  0.3225  0.4400
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   5.8300     0.1756  33.194 1.86e-15 ***
## trt2          0.7400     0.2484   2.979  0.00936 **
## trt3          0.4300     0.2484   1.731  0.10393
```

```
## trt4         -0.6325     0.2484  -2.546  0.02236 *
## trt5         -0.4000     0.2484  -1.610  0.12815
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3513 on 15 degrees of freedom
## Multiple R-squared:  0.7358, Adjusted R-squared:  0.6654
## F-statistic: 10.44 on 4 and 15 DF,  p-value: 0.0003009
```

```r
anova(model_gy)
```

```
## Analysis of Variance Table
##
## Response: gy
##           Df Sum Sq Mean Sq F value    Pr(>F)
## trt        4 5.1551 1.28877  10.445 0.0003009 ***
## Residuals 15 1.8509 0.12339
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

This model evaluates whether seedling density significantly affects grain yield. The ANOVA tests whether mean yield differs across treatments. All statistical analyses were performed at a 5% significance level. The results indicate a statistically significant effect of seedling density on grain yield ($p < 0.05$). Since treatment effects were significant, post-hoc comparisons were conducted to identify differences among treatments.

3.2. Mean Separation (Post-hoc Test)

```r
library(emmeans)
```

```
## Warning: package 'emmeans' was built under R version 4.5.3
```

```
## Welcome to emmeans.
## Caution: You lose important information if you filter this package's results.
## See '? untidy'
```

```r
library(multcomp)
```

```
## Warning: package 'multcomp' was built under R version 4.5.2
```

```
## Loading required package: mvtnorm
```

```
## Warning: package 'mvtnorm' was built under R version 4.5.3
```

```
## Loading required package: survival
```

```
## Loading required package: TH.data
```

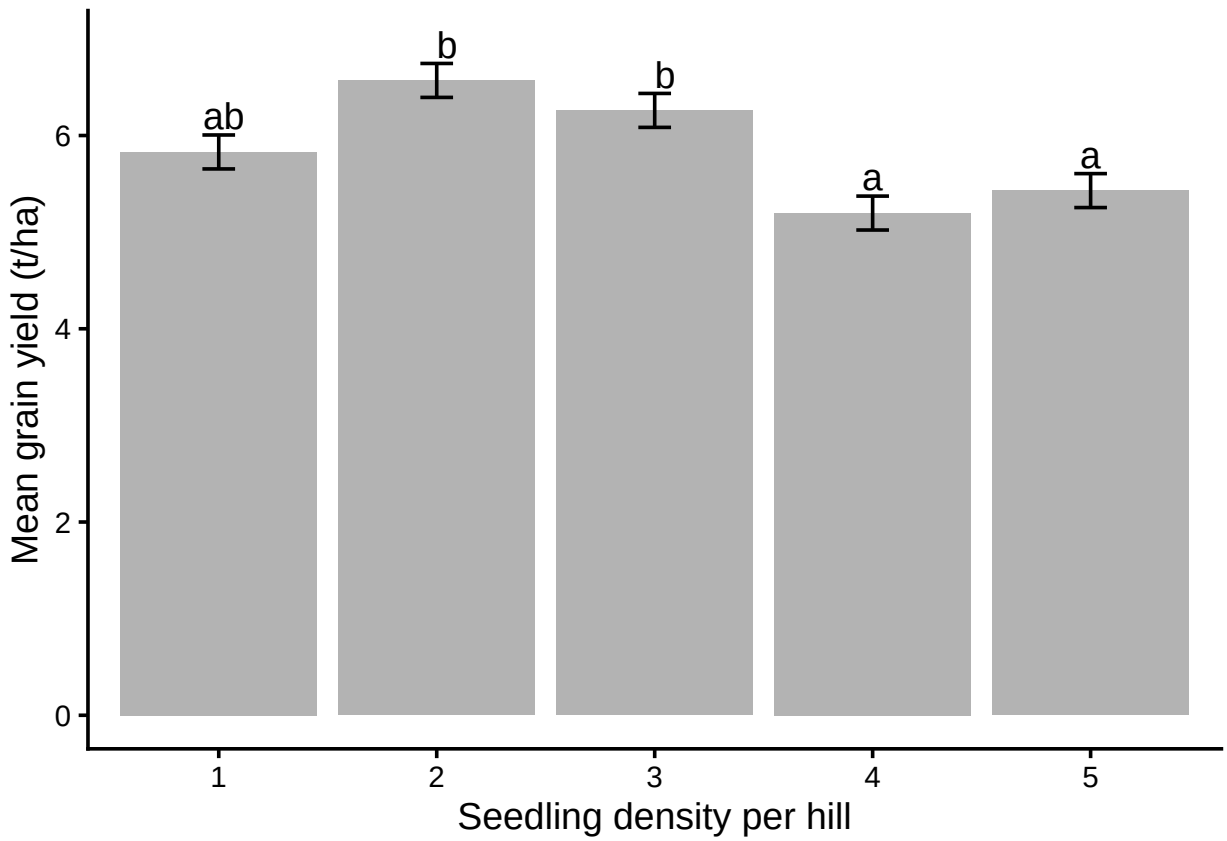
```
## Warning: package 'TH.data' was built under R version 4.5.2
```

```
## Loading required package: MASS

## Warning: package 'MASS' was built under R version 4.5.3

##
## Attaching package: 'MASS'

## The following object is masked from 'package:dplyr':
##
##     select

##
## Attaching package: 'TH.data'

## The following object is masked from 'package:MASS':
##
##     geyser
```

```r
emmeans_gy <- emmeans(model_gy, ~ trt)
pairs(emmeans_gy)
```

```
##  contrast    estimate    SE df t.ratio p.value
##  trt1 - trt2   -0.740 0.248 15  -2.979  0.0611
##  trt1 - trt3   -0.430 0.248 15  -1.731  0.4456
##  trt1 - trt4    0.632 0.248 15   2.546  0.1320
##  trt1 - trt5    0.400 0.248 15   1.610  0.5134
##  trt2 - trt3    0.310 0.248 15   1.248  0.7248
##  trt2 - trt4    1.373 0.248 15   5.526  0.0005
##  trt2 - trt5    1.140 0.248 15   4.590  0.0028
##  trt3 - trt4    1.062 0.248 15   4.278  0.0051
##  trt3 - trt5    0.830 0.248 15   3.342  0.0310
##  trt4 - trt5   -0.233 0.248 15  -0.936  0.8784
##
## P value adjustment: tukey method for comparing a family of 5 estimates
```

```r
cld_gy <- cld(emmeans_gy, Letters = letters)
cld_df <- as.data.frame(cld_gy)

ggplot(cld_df, aes(x = trt, y = emmean)) +
  geom_col(fill = "grey70") +
  geom_errorbar(aes(ymin = emmean - SE, ymax = emmean + SE), width = 0.15) +
  geom_text(aes(label = .group, y = emmean + SE + 0.2), size = 5) +
  labs(x = "Seedling density per hill",
       y = "Mean grain yield (t/ha)") +
  theme_classic(base_size = 14)
```



Post-hoc comparisons were conducted using Tukey-adjusted pairwise tests. Letters above each bar represent significance groupings: Treatments sharing the same letter are not significantly different, whereas treatments with different letters differ significantly at the 5% level.

3.3. Relationship Between Plant Height and Grain Yield

```r
model_ph <- lm(gy ~ ph, data = data)
summary(model_ph)
```

```
##
## Call:
## lm(formula = gy ~ ph, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.99977 -0.40883  0.08934  0.39740  0.79174
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -4.40029    3.74432  -1.175   0.2552
## ph           0.11255    0.04106   2.741   0.0134 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 0.524 on 18 degrees of freedom
## Multiple R-squared:  0.2945, Adjusted R-squared:  0.2553
## F-statistic: 7.513 on 1 and 18 DF,  p-value: 0.01343
```

This regression model examines the relationship between plant height and grain yield.

3.4. Multiple Regression Model

```r
model_multi <- lm(gy ~ ph + nt + et + fg + tgw, data = data)
summary(model_multi)
```

```
##
## Call:
## lm(formula = gy ~ ph + nt + et + fg + tgw, data = data)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.229691 -0.056813 -0.007156  0.070449  0.283978
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.525037   1.671382   1.511  0.15309
## ph          -0.001911   0.015061  -0.127  0.90081
## nt          -0.004575   0.001236  -3.700  0.00238 **
## et           0.021247   0.002110  10.070 8.57e-08 ***
## fg          -0.001476   0.009144  -0.161  0.87408
## tgw         -0.015025   0.022665  -0.663  0.51815
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.158 on 14 degrees of freedom
## Multiple R-squared:  0.9501, Adjusted R-squared:  0.9323
## F-statistic: 53.33 on 5 and 14 DF,  p-value: 1.28e-08
```

This model evaluates the combined effects of key agronomic traits on grain yield.

3.5. RCBD Model (Including Replication)

```r
model_rcbd <- lm(gy ~ trt + rep, data = data)
summary(model_rcbd)
```

```
##
## Call:
## lm(formula = gy ~ trt + rep, data = data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -0.5265 -0.1911 -0.0545  0.2090  0.6155
```

```
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   5.8865     0.2304  25.551 7.86e-12 ***
## trt2          0.7400     0.2576   2.873   0.0140 *
## trt3          0.4300     0.2576   1.669   0.1209
## trt4         -0.6325     0.2576  -2.456   0.0303 *
## trt5         -0.4000     0.2576  -1.553   0.1464
## rep2         -0.2320     0.2304  -1.007   0.3338
## rep3         -0.0700     0.2304  -0.304   0.7664
## rep4          0.0760     0.2304   0.330   0.7472
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3643 on 12 degrees of freedom
## Multiple R-squared:  0.7727, Adjusted R-squared:  0.6402
## F-statistic: 5.829 on 7 and 12 DF,  p-value: 0.004006
```

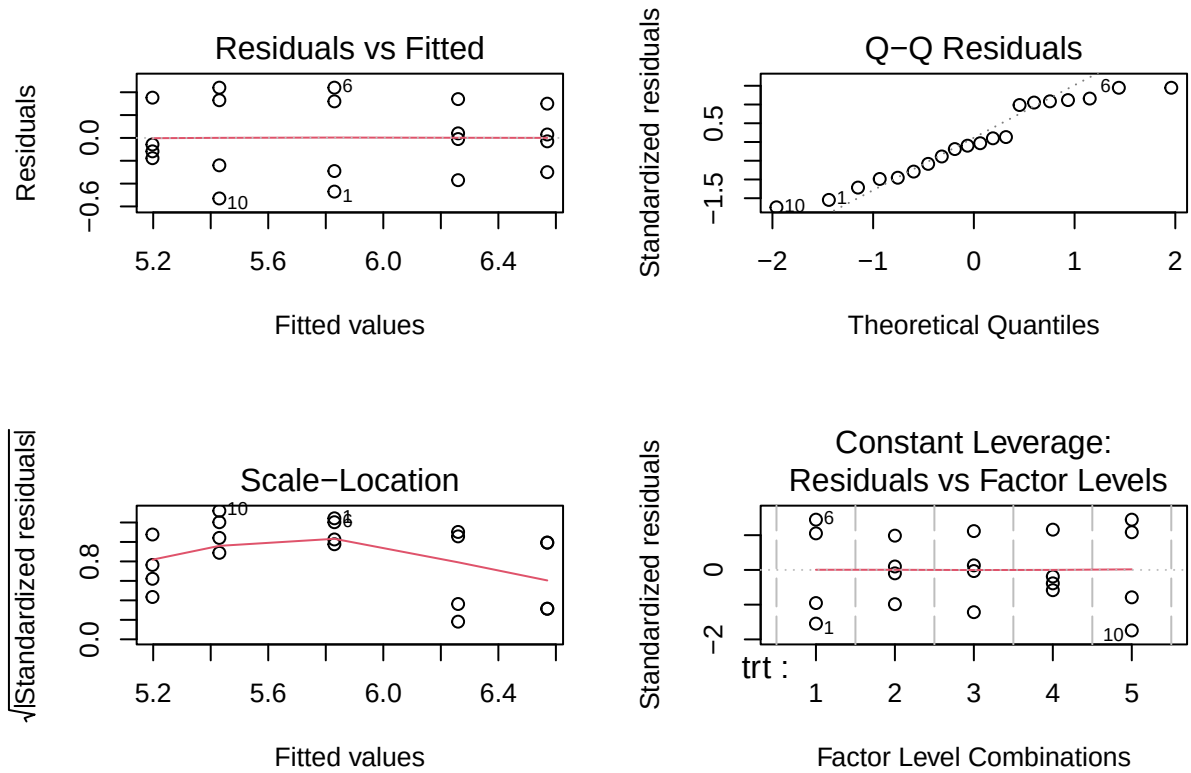
```r
anova(model_rcbd)
```

```
## Analysis of Variance Table
##
## Response: gy
##           Df Sum Sq Mean Sq F value    Pr(>F)
## trt        4 5.1551 1.28877  9.7130 0.0009642 ***
## rep        3 0.2587 0.08622  0.6498 0.5980743
## Residuals 12 1.5922 0.13268
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

This model accounts for the randomized complete block design by including replication as a factor.

4. Model Diagnostics

```r
par(mfrow = c(2,2))
plot(model_gy)
```



Diagnostic plots were used to evaluate model assumptions such as normality and homogeneity of variance. Diagnostic plots indicate that model assumptions of normality, homoscedasticity, and independence are reasonably satisfied, supporting the validity of the linear model results.

5. Reproducible and Efficient Coding

5.1. Data Wrangling

```
data_summary <- data %>%
  group_by(trt) %>%
  summarise(
    mean_gy = mean(gy),
    se_gy = sd(gy)/sqrt(n()),
    mean_nr = mean(nr),
    mean_bc = mean(bc))
data_summary
```

```
## # A tibble: 5 x 5
##   trt   mean_gy se_gy mean_nr mean_bc
##   <fct>   <dbl> <dbl>   <dbl>   <dbl>
## 1 1        5.83 0.224  86572     2.22
## 2 2        6.57 0.123 105672     2.47
## 3 3        6.26 0.146  96422     2.33
```

```
## 4 4        5.20 0.120  66854.    1.91
## 5 5        5.43 0.231  72252     1.97
```

5.2. Function for Linear Models

```r
run_model <- function(response) {
  formula <- as.formula(paste(response, "~ trt"))
  model <- lm(formula, data = data)
  summary(model)
}

run_model("gy")
```

```
##
## Call:
## lm(formula = formula, data = data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -0.5300 -0.2525 -0.0200  0.3225  0.4400
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   5.8300     0.1756  33.194 1.86e-15 ***
## trt2          0.7400     0.2484   2.979  0.00936 **
## trt3          0.4300     0.2484   1.731  0.10393
## trt4         -0.6325     0.2484  -2.546  0.02236 *
## trt5         -0.4000     0.2484  -1.610  0.12815
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3513 on 15 degrees of freedom
## Multiple R-squared:  0.7358, Adjusted R-squared:  0.6654
## F-statistic: 10.44 on 4 and 15 DF,  p-value: 0.0003009
```

```r
run_model("ph")
```

```
##
## Call:
## lm(formula = formula, data = data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.7250 -0.4612  0.0350  0.8044  2.5750
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  93.0250     0.7545 123.289  < 2e-16 ***
## trt2          1.7550     1.0671   1.645 0.120819
## trt3         -1.9400     1.0671  -1.818 0.089076 .
## trt4         -4.2150     1.0671  -3.950 0.001283 **
```

```
## trt5         -5.0375     1.0671  -4.721 0.000273 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.509 on 15 degrees of freedom
## Multiple R-squared:  0.7902, Adjusted R-squared:  0.7343
## F-statistic: 14.13 on 4 and 15 DF,  p-value: 5.667e-05
```

```r
run_model("nt")
```

```
##
## Call:
## lm(formula = formula, data = data)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -38.75 -16.25   3.75  13.75  36.25
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   265.00      11.85  22.363 6.22e-13 ***
## trt2           93.75      16.76   5.594 5.12e-05 ***
## trt3           33.75      16.76   2.014   0.0623 .
## trt4           26.25      16.76   1.566   0.1381
## trt5           48.75      16.76   2.909   0.0108 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 23.7 on 15 degrees of freedom
## Multiple R-squared:  0.6947, Adjusted R-squared:  0.6133
## F-statistic: 8.533 on 4 and 15 DF,  p-value: 0.0008485
```

5.3. Iteration (Loop)

```r
traits <- c("gy", "ph", "nt", "et", "fg")

results <- lapply(traits, function(x) {
  model <- lm(as.formula(paste(x, "~ trt")), data = data)
  anova(model)
})

names(results) <- traits
results
```

```
## $gy
## Analysis of Variance Table
##
## Response: gy
##           Df Sum Sq Mean Sq F value    Pr(>F)
## trt        4 5.1551 1.28877  10.445 0.0003009 ***
## Residuals 15 1.8509 0.12339
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## $ph
## Analysis of Variance Table
##
## Response: ph
##           Df  Sum Sq Mean Sq F value    Pr(>F)
## trt        4 128.692  32.173  14.128 5.667e-05 ***
## Residuals 15  34.158   2.277
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## $nt
## Analysis of Variance Table
##
## Response: nt
##           Df Sum Sq Mean Sq F value    Pr(>F)
## trt        4  19170  4792.5  8.5326 0.0008485 ***
## Residuals 15   8425   561.7
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## $et
## Analysis of Variance Table
##
## Response: et
##           Df  Sum Sq Mean Sq F value    Pr(>F)
## trt        4 15798.1  3949.5  17.448 1.636e-05 ***
## Residuals 15  3395.3   226.4
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## $fg
## Analysis of Variance Table
##
## Response: fg
##           Df Sum Sq Mean Sq F value    Pr(>F)
## trt        4 598.69 149.672  9.8774 0.0004034 ***
## Residuals 15 227.30  15.153
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

5.4. Data Check

```r
if(any(is.na(data))) {
  print("Missing values detected")
} else {
  print("No missing values in dataset")
}
```

```
## [1] "No missing values in dataset"
```

6. Key Findings

- Grain yield varied significantly across seedling density treatments, with intermediate densities (2–3 seedlings per hill) generally producing higher yields.
- Very low or very high seedling densities resulted in comparatively lower yield.
- Plant height showed a positive association with grain yield, indicating the importance of vegetative growth.
- Tiller number and effective tillers contributed to yield variation in the multiple regression model.
- Grain yield was positively related to net return, indicating higher productivity leads to better profitability.
- Inclusion of replication in the RCBD model improved the precision of treatment comparisons.

7. Reproducibility Statement

This analysis was conducted using R and R Markdown. All steps, including data import, visualization, and statistical analysis, are fully reproducible.